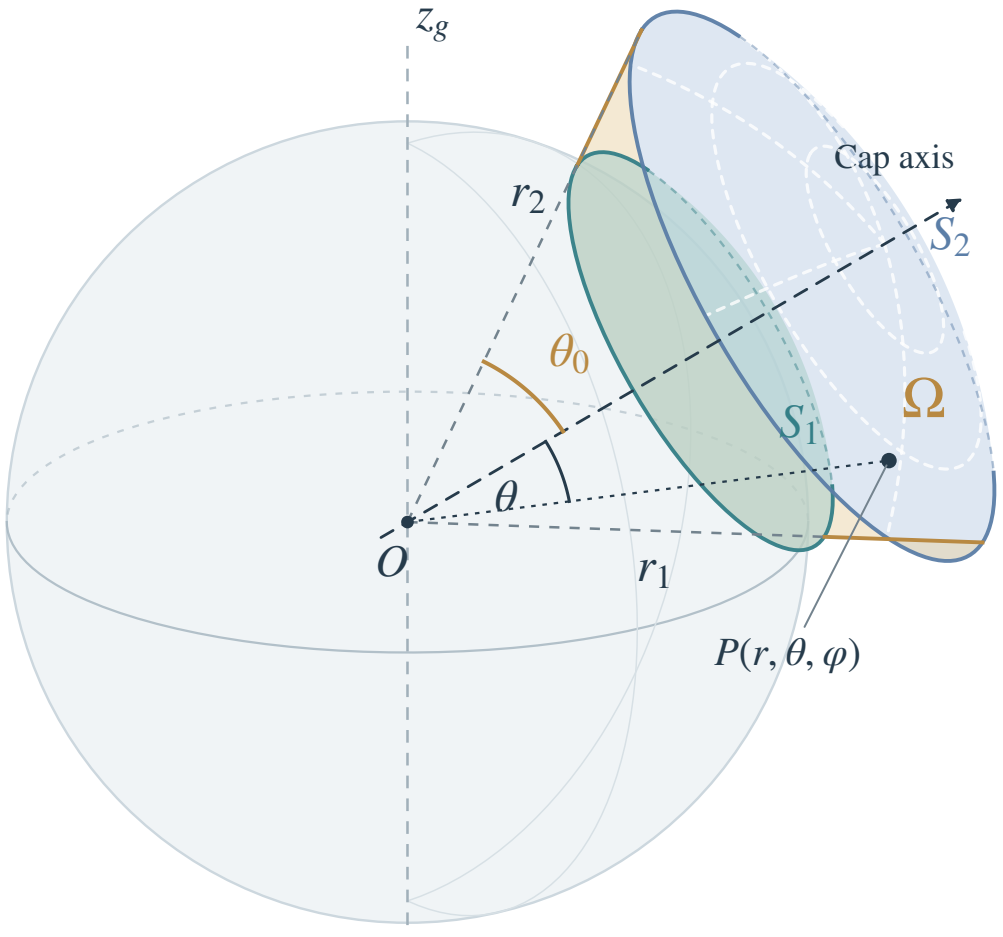
(a) Spherical-conical geometry



$$\Omega: \quad r_1 \le r \le r_2, \quad 0 \le \theta \le \theta_0, \quad 0 \le \varphi < 2\pi$$

(b) Family-specific stable evaluation

Validate input and select the regime
m, θ, θ_0, ν or τ; regular axis / endpoint limits

Legendre
$\lambda = \nu(\nu+1)$
Scaled forms
+ residual atlas
Analytic ∂_ν

Mehler
$\lambda = -(\tau^2 + \frac{1}{4})$
Positive series
or residual atlas
Riccati cap norm

Constant
$\lambda = 0$
Analytic shape
Positive integral
2-D norm atlas

Preserve scale and floating-point range
Component-wise signs, log magnitudes and status

Values and angular derivatives
$L, L_\theta, L_{\theta\theta}$; $\widehat{K}, \widehat{C}$ and derivatives

Residual recovery: (c). Legendre roots and normalization: (d).

(c) Residual compression and recovery

OFFLINE

Generate carrier-removed source fields
Direct kernels; Riccati and positive-series norms

Compress with Chebyshev polynomials
Cell eligibility checks; checksummed atlas

ONLINE

Admissible lookup?

Yes

Nested Clenshaw
Restore carrier
and normalization

No

Non-table route
Family-specific
analytic / direct path

Consistent signed-log output
Report failure if no admissible route exists

Smooth residuals and log norms are tabulated.

(d) Legendre roots and cap normalization

Analytic degree sensitivity
$(L, L_\theta, S, S_\theta), \quad S = \partial_\nu L$

Boundary roots
Action-based predictor
Computed sign bracket
Safeguarded refinement
Public-evaluator residual
and ordering checks

Cap squared norm
$$I = \frac{\sin\theta_0}{2\nu+1} \times (L_\theta S - L S_\theta)_{\theta_0}$$
Integral alternative
when required
Positivity / scale checks

Cap-unit Legendre basis
$\widehat{L} = L/\sqrt{I}, \quad \widehat{L}_\theta, \ \widehat{L}_{\theta\theta}$

R-SCHA field kernels
$V \longrightarrow \mathbf{B} \longrightarrow \mathbf{T}$

Spectral sensitivities link values, roots and cap norms.